

ALT/GPT LIQUIFORM (LABTEST, Brazil)

INTENDED USE

Kinetic system for Alanine Aminotransferase (ALT/GPT) activity determination.

TEST PRINCIPLE

ALT catalyzes the transfer of amino groups of Alanine for α - Ketoglutarate, yielding pyruvate and Glutamate. The pyruvate is reduced to lactate by for action of the lactate dehydrogenase (LDH) which oxidizes NADH to NAD^+ .

The reduction of the absorbance at 340 nm, as consequence of NADH oxidation, must be photometrically monitored, and is proportional to the ALT activity in the sample.

Aminotransferase determination involves the following reactions:



PROVIDED REAGENTS

A. Reagent 1: Store between 2 – 8 °C

TRIS buffer (132.5 mmol/L), L-Alanine (687.5 mmol/L); LDH (≥ 2300 U/L) and sodium azide (0.095%).

B. Reagent 2: Store between 2 – 8 °C

TRIS buffer (20 mmol/L), NADH (1320 $\mu\text{mol/L}$); Ketoglutarate (82.5 mmol/L) and sodium azide (0.095%).

C. Reagent 3: Store between 2 – 8 °C

TRIS buffer (20 mmol/L), pyrooxal phosphate (11.1 mmol/L); sodium azide (0.095%).

Reagents are for “in vitro” diagnostic use. Use the reagents according to the working procedures for clinical laboratories.

SAMPLE -Serum

REQUIRED MATERIAL

- Spectrophotometer.
- Micropipettes and pipettes for measuring the stated volumes
- Water bath at temperature indicated under PROCEDURE.
- Stopwatch.

ASSAY CONDITIONS

- Wavelength: 340 nm
- Reaction temperature: 37°C

MANUAL PROCEDURE

Enzymatic Activity Determination using the pyridoxal phosphate

In order to achieve traceable results to IFCC¹ Procedure, is needed the use of the two-reagent method, to occur the enzyme total activation by the pyridoxal phosphate.

Preparation the reagent: Add 0.300 mL of the Reagent 3 to a bottle of Reagent 1(24 mL) and mix. Stability: 21 days at 2 – 8 °C and 24 hours 15 –25 °C when no chemical or microbial contamination occurs. Optionally, a lower volume of the mixture (Reagent 1 + Reagent 3) may be prepared by using one part of the Reagent 3 to 80 parts of Reagent 1.

Procedure – “In our test the volumes were divided in 5 parts”

1. In a test tube labeled “test” or “Calibrator”, Add 0.800 mL of the mixture Reagent 1 + Reagent 3.
2. Add 0.100 mL of the sample or enzymes calibrator, homogenize and incubate a water- bath at 37 ± 0.2 °C. Wait five minutes. After this incubation it is possible wait until 30 minutes to start the kinetic determination with the addition of the Reagent 2.
3. Perform a water Blank measurement at 340 nm.
4. Add. 0.200 mL of the Reagent 2, homogenize and transfer immediately to a cuvette at 37 ± 0.2 °C. Wait one minute.

5. Measure the initial absorbance (A_1). and start simultaneously the timer. Measure the absorbance again after 2 minutes (A_2).

In order to verify the reaction linearity, it is recommended to measure in 1 minute as well, and check if the difference of absorbance in each minute is constant.

CALCULATIONS

It is a usual procedure calculated the enzymatic activity results using a theoretical factor achieved in reaction optimum conditions, described below:

Wavelength: 340 nm

Cuvette at 37 ± 0.2 °C, 10 mm light path.

Pass band ≤ 2 nm

Sray light $\leq 0.1\%$

If one of the correlated parameters is modified, it is recommended to apply an enzymes calibrator indicated by the reagent manufacturer. Labtest Diagnostica recommends Calibra series to perform the ALT/GPT system calibration.

$$\Delta A/\text{minute}(\text{test or calibrator}) = \frac{A_1 - a_2}{2}$$

$$\text{Factor:} = \frac{\text{Calibrator activity}}{\Delta A/\text{minute}}$$

$$\text{ALT activity (U/L)} = \Delta A/\text{minute (test)} \times \text{Factor}$$

If all the correlated parameters are fulfilled, the theoretical factor (1746) can be applied.

REFERENCE

Reference procedure for the Measurement of catalytic Concentration of Alanine Aminotransferase. Cin Chem Lab Med 2002, 40 (70): 718-24.